

MVBot Day 1: visual guidance basics

Learning objectives:

Navigation is fundamental to the ultimate reason for locomotion. It provides the highest-level control of the general behaviour and direct the motor system to execute appropriate commands. Through the in-class activities and coursework, we hope to achieve the following objectives:

- Understand the general guidance-navigation-control (GNC) framework in robotics
- Appreciate how animals tackle navigation problems
- Learn a brief history of the two-wheel robot system
- Learn the constraints and kinematics model of the two-wheel robot
- Experience some characteristics of a motor through hands-on experiments
- Practice using visual information for basic motor control

In-class activities:

1. Lecture in motor output and navigation [Dr Huai-Ti Lin]

Beside object manipulation, navigation is perhaps the most important topic for sensorimotor control in animals and robots. This lecture will introduce the challenges of navigation and the experimental two-wheel mobile platform.

~~~~ 10min Break ~~~~

### 2. Robot foraging demonstration

### 3. Mobile robot kit distribution

We will distribute the machine vision robot platform for exploring different sensorimotor topics. Pre-assembled units are tested before the class and ready to use. It is your responsibility to maintain all robots' functionality for the duration of the course.

Check your assembled robot against the following criteria (talk to a TA if there is any issue):

- Are both motors properly aligned and secured on the body plate?
- Are both wheels and the casting wheel secured?
- Is the battery properly mounted and secured?
- Are all signal cables neatly folded and secured down?
- Is the SD card present and the USB port access is clear?
- Carefully but firmly rotates the periscope when the robot is off to verify gear meshing.
- With USB connected and the rear switch at the idle mode, do you see a red charging light?
- Locate the power shutdown button under the robot behind the pan servo.

All robot kits are teaching equipment, please take good care of them. 2~3 students might share the guardianship of one robot. Please coordinate within the group to protect the system.

### 4. Test differential drive robot units [hands-on]

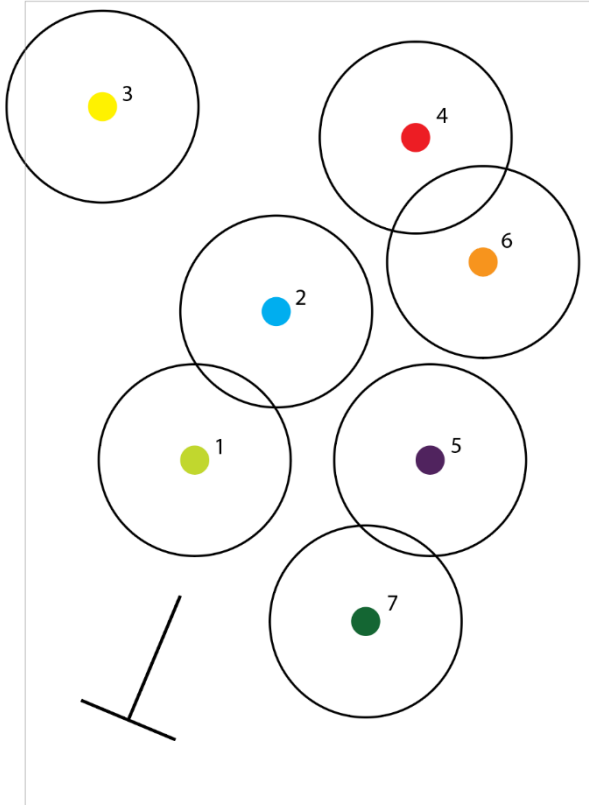
For the hands-on session, please start running some example codes (***servos.py*** and ***camera.py***) from the course repository ([https://github.com/dragonflyneuro/ICL\\_BioEng\\_ALBIR\\_OpenMV\\_V2](https://github.com/dragonflyneuro/ICL_BioEng_ALBIR_OpenMV_V2)) for OpenMV and try out the elements of the custom codebase. Once you have done that you should start the homework exercises "*Visually Guided Foraging*" (see next page for details).

The general robot operation manual can be found in the Github repository.

Refer to "**Getting Started with MVBot\_v3**" for instructions.

## 5: Visually Guided Foraging [hands-on]

To make sure everyone has the basic operational knowledge of the robot platform, your first assignment is to program the robot to locate and go toward a colour patch on the floor. This may be straightforward but you need to go through the process of testing key elements in the codebase. You don't need to perform camera pan for this. The task will be outlined below:



- The robot will start on the starting position (T symbol), wheel axis aligned with the shorter line and the robot pointing along the longer line. (game board is provided). It must go toward the colour patches in the assigned sequence.
- The robot body should point at the colour dot by the time it reaches the 10cm radius (from the visual mirror location) to be considered.
- The reward collection must be done within one continuous autonomous episode. You cannot have one code for one colour and switch code in between.

| Colour patch | Mark |
|--------------|------|
| 1            | 40%  |
| 2            | 55%  |
| 3            | 70%  |
| 4            | 80%  |
| 5            | 90%  |
| 6            | 95%  |
| 7            | 100% |

The assessment will run as follows:

- Colour thresholds for the first 4 patches will be provided – you will have to set the parameters for the rest of the colours for maximum marks.
- We might adjust the organisation of the colour patches to guarantee that you are not performing dead reckoning.

The assessment tests your basic operational knowledge and troubleshooting skills. **You can self-evaluate your ability to operate the robot** within the two-student pair. This will be your first experience of live robot demonstration in this class.